AI, Robots, And The Labor Market: Exposure, Adoption, Displacement, And Transition Through 2040

A governed forecasting and causal-research framework for separating technical capability from realized labor outcomes

> Revision September 26, 2026. Research design with refreshed descriptive evidence. It is not a calibrated job-loss forecast.

Abstract

This paper develops a prospective, falsifiable framework for estimating labor-market changes associated with artificial intelligence and robotics. It rejects the common shortcut that equates technical task exposure with job loss. Instead, it models the path from capability through firm adoption, substitution, deployment lag, demand response, augmentation, new-task creation, and worker reallocation. The report pairs its research design with recommendations for independent ownership, local enterprise and support during job transitions. The recommendations are policy positions, not calibrated employment forecasts.

What this means for your working life

A useful starting question is which parts of your work are changing, and what your employer does with the time saved. A company might handle more work with its existing team, reduce hiring, change responsibilities or cut positions. Those are different outcomes. A task an AI can perform is not a count of people who will lose their jobs.

The same distinction matters for education: preparing someone for a job requires understanding the work employers still need done, including judgment, verification and responsibility. This report offers a way to investigate those changes; it does not rank safe and doomed careers.

Refreshed national labor indicators

Checked September 26, 2026. Reference months differ because release schedules differ. Values below are seasonally adjusted; earnings are nominal. They are descriptive observations, not estimates of AI's effect.

Indicator	Observation	Reference period	Release
Total nonfarm payroll change	+162,000 jobs	August 2026	BLS September 4
Unemployment rate	4.1%	August 2026	BLS September 4
Private nonfarm average hourly earnings	$37.75	August 2026	BLS September 4
Private nonfarm average weekly hours	34.4 hours	August 2026	BLS September 4
Job openings	7.3 million	July 2026	BLS September 1
Hires	5.1 million	July 2026	BLS September 1
Total separations	5.1 million	July 2026	BLS September 1

Indicator	Observation	Reference period	Release
Quits	3.1 million	July 2026	BLS September 1
Layoffs and discharges	1.7 million	July 2026	BLS September 1

Sources: BLS Employment Situation, September 4 release; BLS July JOLTS, September 1 release. JOLTS values are rounded as reported in the release summary; no new ratios are calculated from rounded values. Payroll jobs and the household survey's unemployment rate use different survey populations. The original combined “BLS-CPS-CES-OEWS” employment label is not carried forward as an unambiguous series identifier.

The next JOLTS release is scheduled for September 29 and the next Employment Situation release for October 2. Neither is included here. Monthly estimates can be revised.

Causal bridge: Not ready. National hiring or payroll movements alone cannot identify AI displacement. Strong claims require linked adoption and outcome data, credible comparison groups and checks for other causes.

What measured business AI use tells us

Census's May 26 analysis reports AI use by 19.8% of businesses in the collection period ending May 3, 2026. This is a dated business-use estimate, not a September observation or a share of jobs automated. The survey wording broadened in November 2025 from use in producing goods or services to use in any business function. Comparisons spanning that change need separate treatment.

The original February supplement percentages and 2022 robotics estimate remain in the source snapshot. This refresh does not reproduce them as current national rates or compare scope-coded rows without their denominator definitions. The 19-sector employment-weighted bridge below remains a historical derived result, not a rerun using the new headline observation.

What to watch next

- Track hiring and entry-level opportunities alongside layoffs, wages and hours. A stable national total can hide difficult transitions.
- Ask adopters whether they increased output, changed roles, reduced recruitment or removed positions; record implementation dates and costs.
- Compare similar firms or industries with different adoption timing, including pre-adoption trends and non-AI changes.
- Measure whether affected workers find new work, how long it takes and whether earnings recover.
- Keep robots separate from software AI when measuring installation costs and deployment times.

Recommendations for ownership and job transitions

People should be able to own the tools that make them productive. Our preferred response to automation combines independent ownership, worker participation and protection during job transitions. The aim is to give people income, control and a lasting stake in the economy, rather than require them to rent access to every tool they need to earn a living.

These are the Forge's recommendations. They are not measured findings that robot ownership will replace a specified number of lost jobs. The ranking reflects implementation priorities, not estimated effect sizes. Income support starts immediately when someone loses work; its position below does not make it less urgent.

1 Make productive ownership accessible

Create voluntary routes for individuals and small firms to buy useful equipment, robots and local AI computing systems. Proposed tools include matched ownership grants, fairly priced equipment loans and transparent financing that transfers title. Prioritize displaced workers and viable local businesses that lack collateral. Funding should include installation, training and working capital, not just the purchase price.

Start with equipment that can perform a defined job reliably for paying customers. A cleaning business, groundskeeping service or small fabrication shop could test whether suitable automation improves its economics; these are candidate use cases, not promises about a particular robot. General-purpose robots should qualify only after their capabilities and costs are demonstrated in the intended setting.

An existing financing route is the SBA Microloan program, which offers loans up to $50,000 through intermediaries and permits machinery and equipment purchases. Eligibility, underwriting and terms apply. This is not a robot grant, a promise of approval or necessarily enough funding for an entire operation. Additional ownership grants proposed here would require new funding.

2 Make ownership include practical control

Buying hardware should come with a usable right to operate, maintain, repair, resell and move the business to another supplier. Recommend clear software terms, replaceable components, accessible diagnostics, data export and local operation where technically feasible. Core purchased functionality should not disappear merely because an optional subscription ends. Security updates and safety requirements still matter.

Hardware title alone is insufficient if a vendor controls the machine, the customer list and the only way to obtain work. Examine recurring software, connectivity, maintenance and platform fees before purchase. An open model's license must actually permit the intended use; downloadable weights alone do not establish that right.

A short trial rental can help avoid buying unsuitable equipment. Leasing can sometimes be economical, but it should remain a choice rather than a permanent toll on a person's ability to work. Prefer arrangements with clear costs, an exit and a credible route to ownership when ownership is the objective.

3 Support shared ownership and worker buyouts

Where one person cannot afford a machine, workers or local businesses could jointly own it through a cooperative. Members need defined scheduling rights, voting rights, income allocation, repair reserves and a fair way to leave. A shared workshop can make equipment accessible without requiring each participant to borrow for an underused machine.

Help employees purchase viable local businesses from retiring owners and modernize them together. USDA Cooperative Services describes employee purchases and worker cooperatives as an available business structure. That establishes a practical ownership route, not proof that every conversion succeeds. Other employee-ownership structures can suit some firms, but a financial stake and a voice in decisions are separate things and should be specified explicitly.

4 Help owners find customers and build local services

A machine without customers is a cost. Pair financing with paid pilot contracts, customer discovery, pricing help and introductions to buyers. Recommend smaller, accessible procurement lots where appropriate so qualified local firms can compete. Keep quality, safety and fair competition requirements intact.

Build local work around installation, maintenance, repair, supervision and useful services enabled by the equipment. Support shared workshops and technical colleges as places to test tools and learn their limits. Measure the net effect on nearby firms too: shifting the same customers between businesses is not automatically job creation.

5 Tie paid training to real work

Provide paid, task-specific training linked to an employer, a service contract or an ownership plan. Cover equipment operation, verification, maintenance, sales and basic business finance where relevant. Include apprenticeships and accessible routes for people without advanced degrees. Childcare, transport and training-time income determine whether

many workers can participate.

Measure placement, earnings recovery and durable business income rather than course completions alone. The OECD's work on AI and skills provides background on adapting training to changing work. The specific ownership-linked package here is a Forge proposal.

6 Protect income during the transition

Strengthen access to unemployment support, continued health coverage, job-search help and retraining support. Consider time-limited wage insurance for people accepting lower-paid work, with evaluation of whether it improves recovery or subsidizes persistently low wages. Benefit transitions should not create abrupt penalties for a worker testing a small business.

Nobody should have to take on equipment debt to qualify for help. People who cannot or do not want to operate a business need credible employment and income options. Ownership should widen their choices. It is not a substitute for a safety net.

7 Share productivity gains within existing firms

Encourage meaningful employee ownership, transparent profit sharing and worker participation in how automation changes jobs. Consider shorter hours with protected pay where realized productivity gains can support them. Avoid promising that technology alone can fund every arrangement.

Ownership plans should have understandable rights, fair valuations and protections against excessive concentration of workers' savings in the same business that pays their wages. Profit sharing can distribute income, but it should not be mislabeled as ownership. Training and redeployment should be considered before avoidable layoffs.

8 Keep entry and competition open

Support interoperability, repair access and the ability to change suppliers without losing business records. Public support for automation should include opportunities for smaller firms and ownership by workers, rather than default exclusively to large operators. Apply competition policy to exclusionary conduct on its evidence; company size alone does not establish misconduct.

If labor demand falls broadly despite these measures, test wider income-sharing options, including a publicly accountable capital fund with citizen dividends. Such a proposal needs a funding source, governance rules and evidence about who gains and who pays. Do not present an unfunded dividend as a solution already available.

Test the economics before scaling

For each candidate business, estimate revenue at realistic paid utilization and subtract materials, energy, software, repairs, insurance, finance costs and a replacement reserve. Count human supervision, administration and sales time at a fair wage. Test a low-demand period and a prolonged equipment outage. Owning a robot does not create guaranteed passive income.

A proposed local pilot would first identify paying demand and suitable equipment, then trial the service before committing to larger purchases. Evaluate earnings after all costs, owner control, debt burden, downtime, safety, repeat customers and worker transitions at 3, 6 and 12 months. Compare with ordinary small-business and retraining support where feasible. Include unsuccessful firms and spillovers on competing local workers. The goal is durable income and independence, not the number of robots distributed.

Research Question And Scope

This paper designs a method to estimate plausible labor-market transitions from artificial intelligence and robotics without treating technical capability as realized displacement.

Plain English: A task that a machine could perform is not automatically a job that disappears. Firms must adopt the technology, decide to substitute rather than assist workers, and successfully deploy it before employment changes.

Technical read: The estimand separates task exposure, adoption hazard, substitution propensity, deployment lag, equilibrium demand response, and worker reallocation.

Measurement Architecture

The unit of analysis is an occupation-industry-region-firm-size-horizon cell, with worker-level distributional analysis where public microdata permits.

Plain English: The paper will avoid one dramatic national number. It will show ranges by job type, industry, location, company size, and time horizon.

Technical read: Crosswalk O*NET tasks to SOC occupations and NAICS industries, then join adoption, labor-flow, wage, hours, and establishment-dynamics panels with vintage controls.

AI And Robotics As Separate Channels

Cognitive AI and embodied robotics receive separate exposure and adoption measures before interaction effects are estimated.

Plain English: Software agents can affect office work quickly; robots require hardware, installation, safety approval, and redesigned workplaces. They should not share one clock.

Technical read: Estimate cognitive-task and physical-task channels independently, then test interaction terms in sectors where software orchestration and robotics are complementary.

Causal And Forecast Design

The paper combines descriptive monitoring, causal designs, and prospective scenario forecasts while preserving the boundary between them.

Plain English: A forecast can be useful without pretending it proves why a change happened. Causal claims will require comparison groups and pre-trend tests.

Technical read: Use shift-share and event-study designs where identification is credible; benchmark ML models with rolling-origin holdouts; pre-register scenarios and score matured outcomes.

Distribution And Transition

Aggregate resilience can coexist with severe losses for particular occupations, regions, demographic groups, or firm-size classes.

Plain English: The national total may look stable while specific people and communities experience real disruption. The paper will report both.

Technical read: Estimate heterogeneous effects on employment, openings, wages, hours, layoffs, occupational switching, and nonparticipation with multiple-testing controls.

Forecast Scenarios

Five pre-registered scenario families cover slow augmentation, clerical compression, robot-heavy manufacturing displacement, broad agent adoption, and a productivity-led expansion.

Plain English: The scenarios are structured possibilities, not calibrated odds yet. Their weights must be scored and updated as evidence arrives.

Technical read: Scenario weights are explicitly uncalibrated until a sufficient outcome ledger exists; publish horizon-specific intervals and trigger conditions rather than point forecasts.

Measured Adoption By Sector

The original descriptive bridge matched 19 BTOS sectors to private QCEW employment and wage baselines. Its employment-weighted measured AI adoption was 17.19 percent across matched sectors. This is a descriptive map, not an estimate of AI-caused job or wage changes.

Plain English: AI use is uneven across industries. The retained data can show where adoption is concentrated, but one survey wave cannot tell us whether AI caused employment or wage changes.

Technical read: The bridge maps BTOS sector codes to private QCEW two-digit supersectors, preserves source dates, and blocks causal promotion until multiple stable-definition adoption waves and post-adoption outcomes exist.

Claim Contract

SF-LABOR-CLAIM-6754FC8A56CF

Technical task exposure is not equivalent to job loss; realized displacement depends on adoption, substitution, deployment lag, demand response, and worker or firm adaptation.

- Status: external_context_needs_exact_excerpt_and_model_test
- Horizon: all horizons
- Falsifier: A validated panel shows exposure alone predicts realized employment losses as well as models that include adoption and equilibrium responses.

SF-LABOR-CLAIM-598F253F0F55

Measured business AI adoption already differs across firm sizes and sectors; the paper therefore treats heterogeneous labor effects across functions, industries, regions, and firm sizes as a hypothesis to test rather than an established national-shock result.

- Status: exact_source_support_present_model_test_pending
- Horizon: 1-7 years
- Falsifier: Adoption and labor-flow dispersion remain statistically indistinguishable across sectors, firm sizes, and regions after survey-definition controls.

SF-LABOR-CLAIM-480AF844E971

Near-term labor effects may appear first in hiring, openings, hours, wages, and occupational transitions before they produce large changes in aggregate employment stocks.

- Status: external_context_needs_exact_excerpt_and_model_test
- Horizon: 1-5 years
- Falsifier: High-adoption sectors show no leading changes in labor-flow, hours, wage, or transition measures before aggregate employment adjusts.

SF-LABOR-CLAIM-FDC8ACFDC84F

Generative AI and embodied robotics should be modeled as separate but interacting channels because they expose different task bundles, capital requirements, adoption frictions, and deployment timelines.

- Status: exact_source_support_present_model_test_pending
- Horizon: 1-15 years

- Falsifier: A unified exposure measure consistently outperforms separate cognitive-AI and physical-robot channels out of sample without hiding sector-specific errors.

SF-LABOR-CLAIM-B3202C8FE16E

Productivity gains, lower prices, demand expansion, new tasks, and business formation can offset some direct displacement, so net employment effects must be estimated rather than inferred from automation capability.

- Status: exact_source_support_present_model_test_pending
- Horizon: 4-15 years
- Falsifier: Across validated adoption shocks, product demand, new task creation, and firm formation contribute no measurable offset to direct substitution.

SF-LABOR-CLAIM-16C038869C8F

GenAI exposure is distributionally uneven across worker groups; whether that becomes persistent inequality in realized wages, employment, or transitions remains an empirical question requiring subgroup outcome panels.

- Status: exact_source_support_present_model_test_pending
- Horizon: 1-15 years
- Falsifier: Validated exposure and adoption shocks produce no persistent differential effects across the pre-registered worker, occupation, region, or firm-size groups.

Source Registry

These are original registry records. Their snapshot status is historical and does not mean every source was reopened in this update. Newly checked sources appear in the September sections and the companion fact-check ledger.

- U.S. Census Bureau Survey of Income and Program Participation - descriptive job-to-job occupation transitions using official FINYR2 weights (snapshot_validated)
- U.S. Department of Labor O*NET - occupation tasks, work activities, skills, and detailed-work-activity mappings (snapshot_validated)
- U.S. Census Bureau Business Trends and Outlook Survey - AI adoption by business function, industry, firm size, and state (snapshot_validated)
- U.S. Bureau of Labor Statistics Employment Projections - baseline occupational employment, openings, replacement needs, and projected growth (snapshot_validated)
- U.S. Bureau of Labor Statistics JOLTS - openings, hires, quits, layoffs, and total separations by industry (snapshot_validated)
- U.S. Bureau of Labor Statistics Business Employment Dynamics / QCEW - gross establishment job gains and losses by industry, geography, and firm size (snapshot_validated)
- U.S. Census Bureau Business Dynamics Statistics - firm births, deaths, expansions, contractions, job creation, and destruction (snapshot_validated)
- U.S. Bureau of Labor Statistics - employment, unemployment, hours, earnings, occupation wages, and demographic outcomes (snapshot_validated)
- International Labour Organization - task-level generative-AI exposure and transformation-versus-automation framing (snapshot_validated)
- National Bureau of Economic Research - historical local-labor-market design for industrial robot exposure (snapshot_validated)

- OECD - current cross-country labor-market evidence and policy context (snapshot_validated)
- U.S. Census Bureau Annual Business Survey - measured AI and robotics adoption, motives, and reported workforce effects (snapshot_validated)
- National Bureau of Economic Research - workplace productivity, retention, and heterogeneous worker effects from generative AI (snapshot_validated)
- National Bureau of Economic Research - displacement and reinstatement channels from automation and new-task creation (snapshot_validated)
- National Bureau of Economic Research - long-run evidence on new occupational specialties and labor demand (snapshot_validated)
- National Bureau of Economic Research - national generative-AI work adoption, intensity, and occupation or education heterogeneity (snapshot_validated)
- U.S. Bureau of Labor Statistics - aggregate hires level used as a descriptive labor-flow indicator (snapshot_validated)
- U.S. Bureau of Labor Statistics - aggregate quits level used as a descriptive worker-mobility indicator (snapshot_validated)
- U.S. Bureau of Labor Statistics - aggregate layoffs and discharges used as a descriptive displacement indicator (snapshot_validated)
- U.S. Bureau of Labor Statistics - aggregate total separations used as a descriptive labor-flow indicator (snapshot_validated)
- U.S. Bureau of Labor Statistics - total-private employment stock compatible with private-sector hours and earnings (snapshot_validated)
- U.S. Bureau of Labor Statistics - total-private average weekly hours used as a descriptive intensive-margin indicator (snapshot_validated)
- U.S. Bureau of Labor Statistics - total-private average hourly earnings used as a descriptive wage indicator (snapshot_validated)
- U.S. Census Bureau Business Trends and Outlook Survey - national AI-use estimates from the 2026 BTOS AI supplement (snapshot_validated)
- U.S. Census Bureau Business Trends and Outlook Survey - sector AI-use estimates from the 2026 BTOS AI supplement (snapshot_validated)
- U.S. Census Bureau Business Trends and Outlook Survey - employment-size AI-use estimates from the 2026 BTOS AI supplement (snapshot_validated)
- National Center for Science and Engineering Statistics / U.S. Census Bureau Annual Business Survey - 2022 robotics use as a production technology by industry (snapshot_validated)
- National Center for Science and Engineering Statistics / U.S. Census Bureau Annual Business Survey - 2022 robotics use as a production technology by company employment size (snapshot_validated)

Source Acquisition Plan

1. **BLS JOLTS full industry series** - Measure labor-flow changes before relying on aggregate employment stocks. Acceptance: Versioned monthly industry series with release dates, seasonal-adjustment flags, and exact series identifiers.

2. **BLS CES hours and earnings** - Detect intensive-margin adjustment that can precede headcount changes. Acceptance: Versioned monthly industry panel with compatible NAICS vintages and revision metadata.

3. **Census BTOS AI adoption estimates** - Replace exposure-only inference with measured deployment behavior. Acceptance: Public estimate tables segmented at the November 2025 question-definition break.

4. **Census ABS technology and robotics tables** - Keep embodied robotics separate from cognitive AI while testing interaction effects. Acceptance: Public table cells with survey year, industry, firm-size coverage, suppression flags, and definition notes.

5. **O*NET task data and SOC/NAICS crosswalks** - Create version-bound occupation and task exposure features without invented joins. Acceptance: Hashed release files and explicit crosswalk versions with many-to-many mapping weights.

6. **CPS or SIPP worker-transition evidence** - Measure who moves, exits, loses hours, or recovers after adoption shocks. Acceptance: Documented longitudinal linkage or validated transition aggregates with disclosure-safe subgroup coverage.

Research Status

The following scientific work remains to establish causal effects and calibrated forecasts. The policy recommendations express the Forge's position and can be discussed without pretending that their effects have already been measured.

- Status: research_design_not_forecast_ready
- Readiness score: 0.47
- Missing: SOC/NAICS crosswalk aligned to the versioned O*NET snapshot
- Missing: multi-wave BTOS adoption history segmented at the November 2025 definition break
- Missing: multi-wave industry and occupation labor-flow, hours, wage, and transition panels
- Missing: pre-trend, placebo, and control-group diagnostics
- Missing: matured forecast outcome scoring